

Lab 8: Scenario Discovery

CEVE 421/521

2026-03-27

1 Overview

In Lab 7 you evaluated house elevation decisions against 10,000 SLR trajectories and computed robustness metrics. Those metrics ranked decisions across scenarios. They did not identify **which conditions** led to failure.

In this lab you'll use **scenario discovery** to answer a different question:

Under what conditions does my chosen strategy **fail**?

You'll work with the Eijgenraam dike heightening model — a cost-benefit framework for setting flood protection standards in the Netherlands [1]. You'll run the model across 1,000 uncertain futures, label each as success or failure, and use **PRIM** (Patient Rule Induction Method) to find the combinations of uncertainties that drive failure [2].

The key insight: the answer depends on how you define “failure.”

References:

- C. Eijgenraam *et al.* [1] — Eijgenraam dike model (the case study)
- J. H. Friedman and N. I. Fisher [2] — PRIM algorithm

2 Before Lab

Complete these steps **BEFORE** coming to lab:

1. **Accept the GitHub Classroom assignment** (link on Canvas)
2. **Clone your repository:**

```
git clone https://github.com/CEVE-421-521/lab-08-S26-yourusername.git
cd lab-08-S26-yourusername
```

3. **Open the notebook** in VS Code or Quarto preview
4. **Run the first code cell** — it will install any missing packages (may take 10-15 minutes the first time)

! Important

Submission: You will render this notebook to PDF and submit the PDF to Canvas. See Section 7 for details.

3 Setup

Run the cells in this section to load packages, the model, and generate the ensemble. You don't need to modify any of this code — just run it and read the comments.

3.1 Packages

```
# Install SimOptDecisions if needed
try
    using SimOptDecisions
catch
    import Pkg
    try
        Pkg.rm("SimOptDecisions")
    catch
    end
    Pkg.add(; url="https://github.com/dossgollin-lab/SimOptDecisions")
    Pkg.instantiate()
    using SimOptDecisions
end
```

```
if !isfile("Manifest.toml")
    import Pkg
    Pkg.instantiate()
end
```

```
using CairoMakie
using DataFrames
using Distributions
using Random
using Statistics
```

```
include("eigenraam.jl")
include("prim.jl")
```

```
Random.seed!(2026)
```

3.2 The Eijgenraam Dike Model

The Eijgenraam model [1] computes the total societal cost of flood protection for a dike ring in the Netherlands. It balances two competing forces:

- **Investment cost:** building the dike higher is expensive, and costs grow exponentially with height
- **Expected flood losses:** without protection, losses grow over time due to sea level rise and economic growth

The annual **flood probability** at year t is:

$$P(t) = P_0 \exp(\alpha \eta t - \alpha H(t))$$

where P_0 is the initial flood probability, α controls how sensitive flood probability is to water level, η is the rate of water level rise, and $H(t)$ is the cumulative dike heightening at time t .

i Note

Notice that α appears **twice**: it makes flood probability grow faster with rising water ($\alpha \eta t$), but it also makes each cm of dike height more effective ($-\alpha H$). This means α 's effect on failure is **non-monotonic** — something scenario discovery can reveal.

3.3 Policy: When and How Much to Heighten

Our policy has two levers:

- **When** to heighten the dike (t_h years from now)
- **How much** to heighten it (u cm)

There's a genuine tradeoff: deferring the investment saves money (discounting), but the dike ring is unprotected while we wait.

3.4 Generate the Ensemble

We sample 1,000 scenarios by drawing each uncertain parameter uniformly within plausible bounds. This gives a well-spread input space — ideal for PRIM.

```
config = EijgenraamConfig() # ring 15 defaults

n_scenarios = 1000
scenarios = map(1:n_scenarios) do _
    EijgenraamScenario(
        P0=rand(Uniform(0.0005, 0.004)),
        alpha=rand(Uniform(0.025, 0.075)),
        eta=rand(Uniform(0.4, 1.2)),
        V0=rand(Uniform(5000.0, 25000.0)),
        delta=rand(Uniform(0.02, 0.06)),
    )
end
```

4 Act 1: Explore the Model

4.1 Run the Ensemble

We fix a policy — heighten the dike by 100 cm immediately (year 0) — and run all 1,000 scenarios through the model.

```
policy = EijgenraamPolicy(; t_heighten=0.0, u_heighten=100.0)
results = explore(config, scenarios, [policy])
```

4.2 Failure Probability Trajectories

The plot below shows how flood probability evolves over time for each scenario. Each line is one scenario. The dashed red line is the current legal safety standard (1/2,000 per year).

```
let
    T = config.T
    max_Pf = config.max_Pf
    t_h = SimOptDecisions.value(policy.t_heighten)
    u = SimOptDecisions.value(policy.u_heighten)
    years = 0:T

    fig = Figure(; size=(750, 400))
    ax = Axis(
        fig[1, 1];
        xlabel="Year",
        ylabel="Failure Probability (1/year)",
        title="Flood Probability Over Time",
        yscale=log10,
        limits=(nothing, (1e-6, 1e-1)),
    )

    # Plot each trajectory
    max_pf_vals = Array(results.max_failure_prob)[1, :]
    for (i, s) in enumerate(scenarios)
        P0 = SimOptDecisions.value(s.P0)
        alpha = SimOptDecisions.value(s.alpha)
        eta = SimOptDecisions.value(s.eta)
        traj = failure_prob_trajectory(P0, alpha, eta, u, t_h, T)

        exceeds = max_pf_vals[i] > max_Pf
        color = exceeds ? (:red, 0.08) : (:gray, 0.05)
        lines!(ax, collect(years), traj; color=color, linewidth=0.5)
    end

    # Safety standard
    hlines!(ax, [max_Pf]; color=:red, linestyle=:dash, linewidth=2, label="Safety")
```

```

standard (1/2000)")
    axislegend(ax; position=:lb)
    fig
end

```

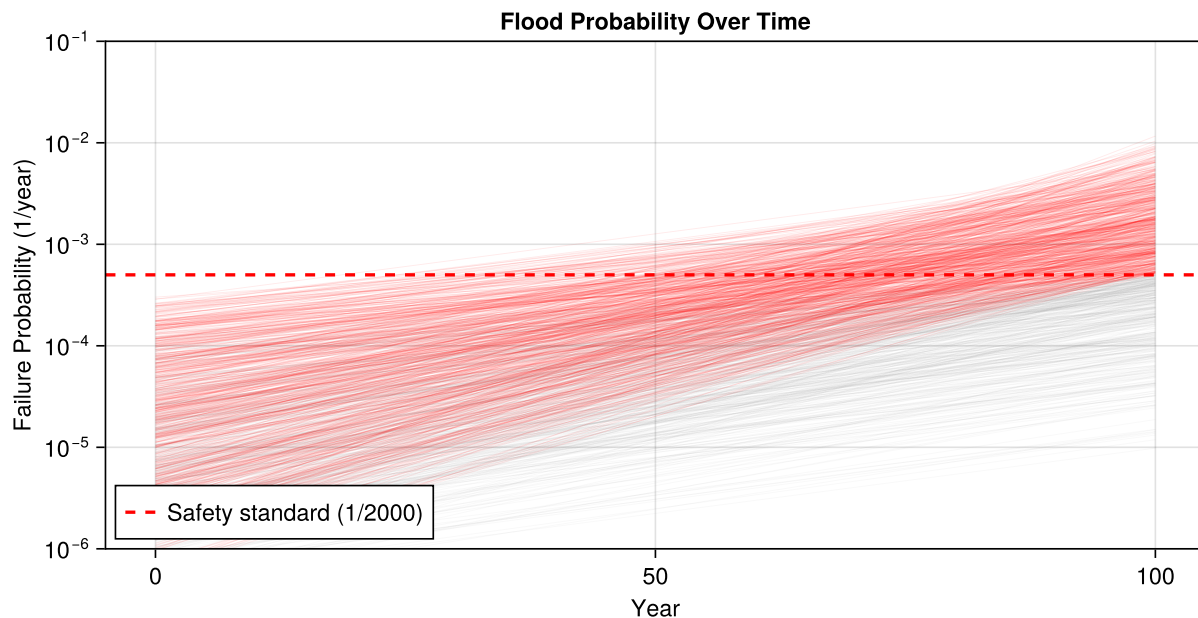


Figure 1: Failure probability trajectories for 1,000 scenarios under the default policy (heighten 100 cm at year 0). Each gray line is one scenario. Lines exceeding the safety standard (dashed red) represent policy failures.

4.3 Your Turn: Choose a Policy

Your Response — Question 1

Choose a different policy (change the timing, the height, or both). Run the ensemble and plot the failure probability trajectories. In 2-3 sentences: why did you choose this policy, and how do the results differ from the default?

Your code here

Your code here

Figure 2

5 Act 2: Scenario Discovery

Now we use the default policy results to ask: **under what conditions does this policy fail?**

5.1 Step 1: Label Success and Failure

We define failure as the maximum flood probability exceeding the legal safety standard (1/2,000 per year). This is the same standard used in Dutch law for coastal dike rings.

```
max_pf = Array(results.max_failure_prob)[1, :] # extract from YAXArray
failures = max_pf .> config.max_Pf
n_fail = sum(failures)
println("${n_fail} of ${n_scenarios} scenarios are failures (${round(100 * n_fail /
n_scenarios; digits=1)})%")
```

```
625 of 1000 scenarios are failures (62.5%)
```

5.2 Step 2: Build the Input Matrix

To run PRIM, we need the scenario parameters as a plain matrix. Each row is one scenario, each column is one uncertain input.

```
param_names = ["P₀", "α", "η", "V₀", "δ"]
X = hcat(
    [SimOptDecisions.value(s.P0) for s in scenarios],
    [SimOptDecisions.value(s.alpha) for s in scenarios],
    [SimOptDecisions.value(s.eta) for s in scenarios],
    [SimOptDecisions.value(s.V0) for s in scenarios],
    [SimOptDecisions.value(s.delta) for s in scenarios],
)
```

5.3 Step 3: Run PRIM

PRIM peels away slices of the input space to find a “box” where failures concentrate. It traces a path from high coverage (large box, catches most failures) to high density (small box, almost all scenarios in the box are failures).

```
trajectory = prim_peel(X, failures)
prim_summary(trajectory, param_names)
```

Step	Coverage	Density	Support	Restricted dimensions
----	-----	-----	-----	-----
1	100.0%	62.5%	1000	(full space)
2	99.5%	65.5%	950	$\eta \geq 0.435$
3	99.4%	68.8%	903	$\eta \geq 0.4744$
4	98.6%	71.8%	858	$\eta \geq 0.5139$
5	97.4%	74.6%	816	$\eta \geq 0.5512$
6	96.3%	77.6%	776	$\eta \geq 0.5803$
7	94.4%	79.9%	738	$\eta \geq 0.6159$
8	92.3%	82.2%	702	$P_0 \geq 0.0007, \eta \geq 0.6159$

9		89.8%		84.1%		667		$P_0 \geq 0.0009$, $\eta \geq 0.6159$
10		87.2%		86.0%		634		$P_0 \geq 0.0009$, $\eta \geq 0.6407$
11		85.0%		88.1%		603		$P_0 \geq 0.0009$, $\eta \geq 0.667$
12		83.2%		90.8%		573		$P_0 \geq 0.0009$, $\eta \geq 0.6944$
13		80.6%		92.5%		545		$P_0 \geq 0.0009$, $\eta \geq 0.7162$
14		78.4%		94.6%		518		$P_0 \geq 0.0009$, $\eta \geq 0.7413$
15		75.8%		96.1%		493		$P_0 \geq 0.0009$, $\eta \geq 0.7644$
16		73.0%		97.2%		469		$P_0 \geq 0.0011$, $\eta \geq 0.7644$
17		70.2%		98.4%		446		$P_0 \geq 0.0011$, $\alpha \leq 0.0728$, $\eta \geq 0.7644$
18		67.2%		99.1%		424		$P_0 \geq 0.0011$, $\alpha \leq 0.0708$, $\eta \geq 0.7644$
19		64.0%		99.3%		403		$P_0 \geq 0.0012$, $\alpha \leq 0.0708$, $\eta \geq 0.7644$
20		61.0%		99.5%		383		$P_0 \geq 0.0014$, $\alpha \leq 0.0708$, $\eta \geq 0.7644$
21		58.1%		99.7%		364		$P_0 \geq 0.0015$, $\alpha \leq 0.0708$, $\eta \geq 0.7644$
22		55.4%		100.0%		346		$P_0 \geq 0.0016$, $\alpha \leq 0.0708$, $\eta \geq 0.7644$

```

let
    coverages = [b.coverage for b in trajectory]
    densities = [b.density for b in trajectory]

    fig = Figure(; size=(600, 400))
    ax = Axis(
        fig[1, 1];
        xlabel="Density (fraction of box that are failures)",
        ylabel="Coverage (fraction of all failures captured)",
        title="PRIM Peeling Trajectory",
    )
    scatterlines!(ax, densities, coverages; markersize=8, linewidth=2)

    # Label first and last
    text!(ax, densities[1], coverages[1]; text="Start", align=:right, :bottom,
    fontsize=12)
    text!(
        ax,
        densities[end],
        coverages[end];
        text="End",
        align=:left, :top,
        fontsize=12,
    )
    fig
end

```

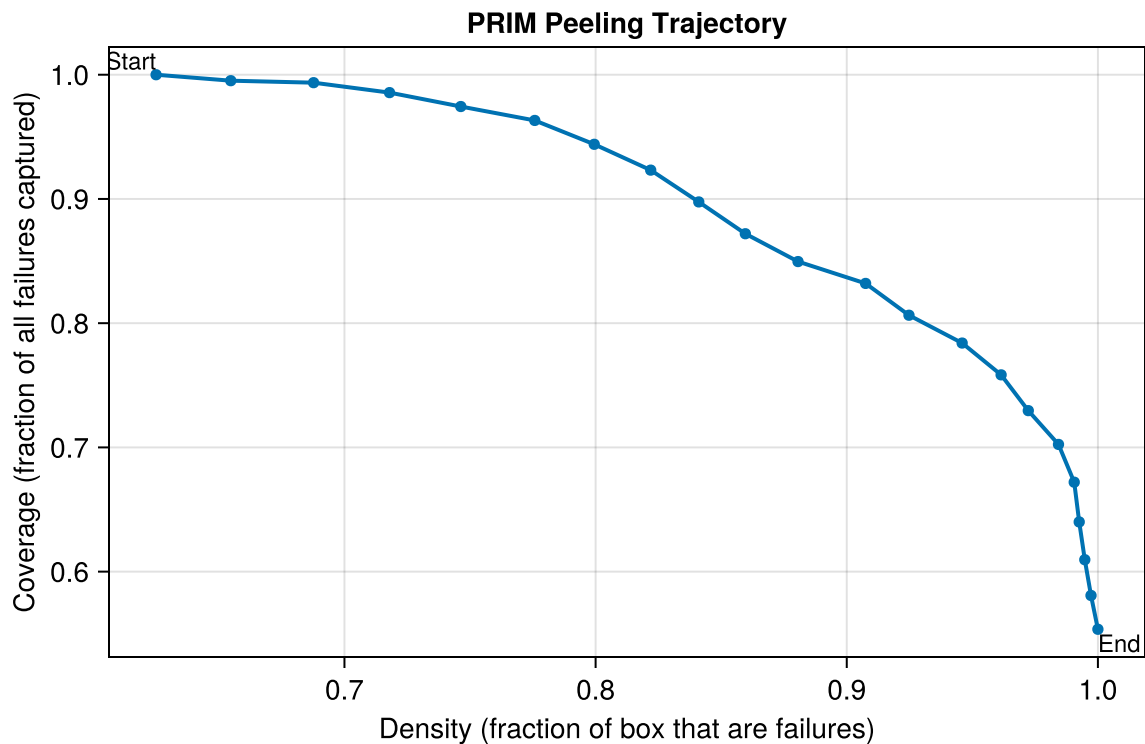


Figure 3: PRIM peeling trajectory: each point is a box. Moving right increases density (precision); moving down decreases coverage (recall). The analyst chooses a box that balances both.

5.4 Step 4: Interpret a PRIM Box

Pick a box from the trajectory that you think gives a good balance of coverage and density. A good starting point is a box with density above 70% and coverage above 50%.

```
# Find a box with density > 0.7 (or the last box if none qualify)
good_idx = findfirst(b -> b.density > 0.7, trajectory)
if good_idx === nothing
    good_idx = length(trajectory)
end
box = trajectory[good_idx]
println("Selected box $(good_idx):")
println(" Coverage: $(round(100 * box.coverage; digits=1))%")
println(" Density:  $(round(100 * box.density; digits=1))%")
println(" Support:  $(box.support) scenarios")
println()
initial = trajectory[1]
for (j, name) in enumerate(param_names)
    lo_changed = box.lower[j] > initial.lower[j]
    hi_changed = box.upper[j] < initial.upper[j]
    if lo_changed || hi_changed
        println(" $name: [$(round(box.lower[j]; sigdigits=3)), $(round(box.upper[j];
```



```
sigdigits=3))])")
    end
end
```

Selected box 4:

Coverage: 98.6%
 Density: 71.8%
 Support: 858 scenarios

η : [0.514, 1.2]

Your Response — Question 2

Look at the PRIM box above. Complete this sentence: "The dike heightening policy fails when ____ AND ____." Use the parameter names and bounds from the box.

Which parameters does PRIM restrict? Which parameters does it leave unrestricted? Does this match your intuition from the failure probability formula?

5.5 Step 5: Change the Failure Definition

Now let's define failure differently. Instead of "does the flood probability exceed the safety standard?", ask: "does the **total cost** (investment + expected losses) exceed a threshold?"

First, look at the distribution of total costs to pick a reasonable threshold.

```
let
    costs = Array(results.total_cost)[1, :]

    fig = Figure(; size=(600, 350))
    ax = Axis(fig[1, 1]; xlabel="Total Cost (M€)", ylabel="Count", title="Distribution
of Total Cost")
    hist!(ax, costs; bins=50, color=(steelblue, 0.7), strokewidth=0.5,
strokecolor=:gray)
    vlines!(ax, [median(costs)]; color=:red, linestyle=:dash, linewidth=2,
label="Median")
    axislegend(ax; position=:rt)
    fig
end
```

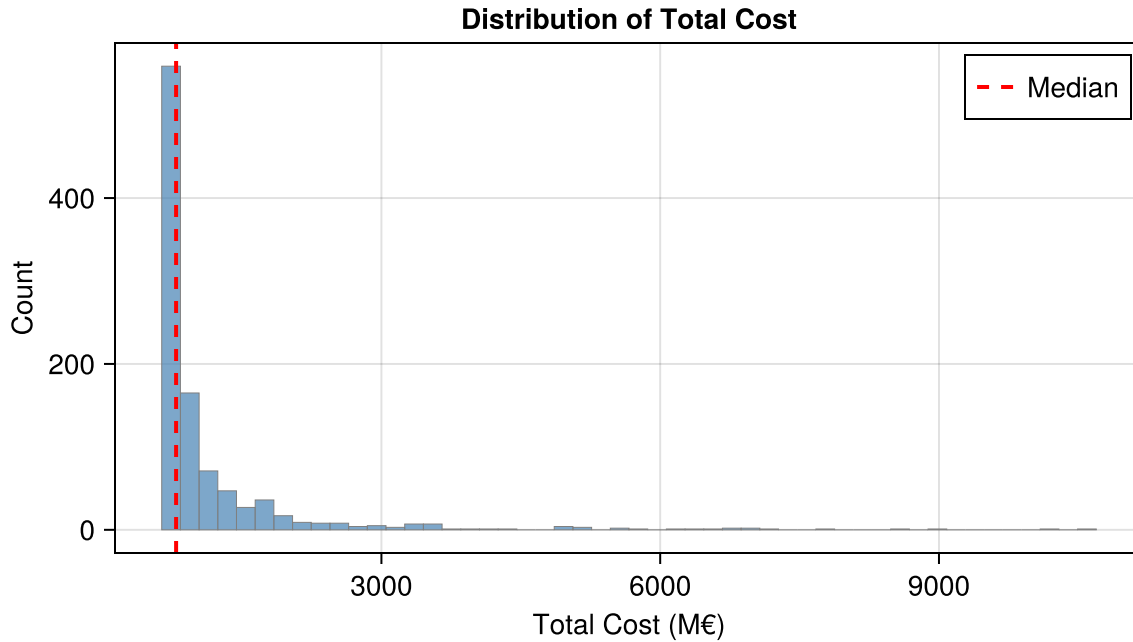


Figure 4: Distribution of total cost across 1,000 scenarios. The dashed line shows the median, which we use as the failure threshold.

Now label failures using the cost-based definition and re-run PRIM.

```
total_costs = Array(results.total_cost)[1, :]
cost_threshold = median(total_costs)
cost_failures = total_costs .> cost_threshold
println("Cost-based failures: $(sum(cost_failures)) of $(n_scenarios) (threshold: $(round(cost_threshold; digits=1)) M€)")
```

```
Cost-based failures: 500 of 1000 (threshold: 790.6 M€)
```

```
cost_trajectory = prim_peel(X, cost_failures)
prim_summary(cost_trajectory, param_names)
```

Step	Coverage	Density	Support	Restricted dimensions
1	100.0%	50.0%	1000	(full space)
2	98.8%	52.0%	950	$\eta \geq 0.435$
3	97.2%	53.8%	903	$P_0 \geq 0.0007, \eta \geq 0.435$
4	95.6%	55.7%	858	$P_0 \geq 0.0007, \eta \geq 0.435, V_0 \geq 6066.1829$
5	94.2%	57.7%	816	$P_0 \geq 0.0007, \eta \geq 0.435, V_0 \geq 6066.1829, \delta \leq 0.0584$
6	92.4%	59.5%	776	$P_0 \geq 0.0007, \eta \geq 0.4737, V_0 \geq 6066.1829, \delta \leq 0.0584$
7	90.4%	61.2%	738	$P_0 \geq 0.0009, \eta \geq 0.4737, V_0 \geq 6066.1829, \delta \leq 0.0584$

8 88.2% 62.8% 6066.1829, $\delta \leq 0.0584$	702 $P_0 \geq 0.0009$, $\alpha \leq 0.0731$, $\eta \geq 0.4737$, $V_0 \geq$
9 85.8% 64.3% 6066.1829, $\delta \leq 0.0584$	667 $P_0 \geq 0.0009$, $\alpha \leq 0.0731$, $\eta \geq 0.5098$, $V_0 \geq$
10 83.8% 66.1% 6066.1829, $\delta \leq 0.0584$	634 $P_0 \geq 0.0009$, $\alpha \leq 0.0731$, $\eta \geq 0.5547$, $V_0 \geq$
11 81.4% 67.5% 6066.1829, $\delta \leq 0.0584$	603 $P_0 \geq 0.0009$, $\alpha \leq 0.0731$, $\eta \geq 0.5879$, $V_0 \geq$
12 79.0% 68.9% 6066.1829, $\delta \leq 0.0568$	573 $P_0 \geq 0.0009$, $\alpha \leq 0.0731$, $\eta \geq 0.5879$, $V_0 \geq$
13 77.8% 71.4% 6066.1829, $\delta \leq 0.055$	545 $P_0 \geq 0.0009$, $\alpha \leq 0.0731$, $\eta \geq 0.5879$, $V_0 \geq$
14 76.2% 73.6% 6066.1829, $\delta \leq 0.0536$	518 $P_0 \geq 0.0009$, $\alpha \leq 0.0731$, $\eta \geq 0.5879$, $V_0 \geq$
15 74.2% 75.3% 6066.1829, $\delta \leq 0.0519$	493 $P_0 \geq 0.0009$, $\alpha \leq 0.0731$, $\eta \geq 0.5879$, $V_0 \geq$
16 71.4% 76.1% 6066.1829, $\delta \leq 0.0519$	469 $P_0 \geq 0.001$, $\alpha \leq 0.0731$, $\eta \geq 0.5879$, $V_0 \geq 6066.1829$,
17 68.6% 76.9% 6066.1829, $\delta \leq 0.0519$	446 $P_0 \geq 0.001$, $\alpha \leq 0.0731$, $\eta \geq 0.5879$, $V_0 \geq 6913.9267$,
18 66.2% 78.1% 6066.1829, $\delta \leq 0.0519$	424 $P_0 \geq 0.001$, $\alpha \leq 0.0731$, $\eta \geq 0.5879$, $V_0 \geq 7686.1245$,
19 63.4% 78.7% 7686.1245, $\delta \leq 0.0519$	403 $P_0 \geq 0.0012$, $\alpha \leq 0.0731$, $\eta \geq 0.5879$, $V_0 \geq$
20 60.6% 79.1% 7686.1245, $\delta \leq 0.0519$	383 $P_0 \geq 0.0013$, $\alpha \leq 0.0731$, $\eta \geq 0.5879$, $V_0 \geq$
21 58.2% 79.9% 7686.1245, $\delta \leq 0.0519$	364 $P_0 \geq 0.0015$, $\alpha \leq 0.0731$, $\eta \geq 0.5879$, $V_0 \geq$
22 55.8% 80.6% 8531.9182, $\delta \leq 0.0519$	346 $P_0 \geq 0.0015$, $\alpha \leq 0.0731$, $\eta \geq 0.5879$, $V_0 \geq$
23 54.0% 82.1% 9264.5363, $\delta \leq 0.0519$	329 $P_0 \geq 0.0015$, $\alpha \leq 0.0731$, $\eta \geq 0.5879$, $V_0 \geq$
24 52.2% 83.4% 10178.1503, $\delta \leq 0.0519$	313 $P_0 \geq 0.0015$, $\alpha \leq 0.0731$, $\eta \geq 0.5879$, $V_0 \geq$
25 50.2% 84.2% 10178.1503, $\delta \leq 0.0519$	298 $P_0 \geq 0.0015$, $\alpha \leq 0.0731$, $\eta \geq 0.6185$, $V_0 \geq$
26 48.0% 84.5% 10178.1503, $\delta \leq 0.0519$	284 $P_0 \geq 0.0016$, $\alpha \leq 0.0731$, $\eta \geq 0.6185$, $V_0 \geq$
27 46.0% 85.2% 10178.1503, $\delta \leq 0.0519$	270 $P_0 \geq 0.0018$, $\alpha \leq 0.0731$, $\eta \geq 0.6185$, $V_0 \geq$
28 43.8% 85.2% 10178.1503, $\delta \leq 0.0519$	257 $P_0 \geq 0.0019$, $\alpha \leq 0.0731$, $\eta \geq 0.6185$, $V_0 \geq$
29 41.8% 85.3% 10178.1503, $\delta \leq 0.0519$	245 $P_0 \geq 0.002$, $\alpha \leq 0.0731$, $\eta \geq 0.6185$, $V_0 \geq 10178.1503$,
30 39.8% 85.4% 10178.1503, $\delta \leq 0.0519$	233 $P_0 \geq 0.002$, $\alpha \leq 0.0731$, $\eta \geq 0.6315$, $V_0 \geq 10178.1503$,

31	38.0%	85.6%	222	$P_0 \geq 0.002, \alpha \leq 0.0731, \eta \geq 0.6666, V_0 \geq 10178.1503,$ $\delta \leq 0.0519$
32	36.6%	86.7%	211	$P_0 \geq 0.002, \alpha \leq 0.0731, \eta \geq 0.6881, V_0 \geq 10178.1503,$ $\delta \leq 0.0519$
33	35.4%	88.1%	201	$P_0 \geq 0.002, \alpha \leq 0.0731, \eta \geq 0.7068, V_0 \geq 10178.1503,$ $\delta \leq 0.0519$
34	34.6%	90.6%	191	$P_0 \geq 0.002, \alpha \leq 0.0731, \eta \geq 0.7221, V_0 \geq 10178.1503,$ $\delta \leq 0.0519$
35	33.4%	91.8%	182	$P_0 \geq 0.002, \alpha \leq 0.0731, \eta \geq 0.7512, V_0 \geq 10178.1503,$ $\delta \leq 0.0519$
36	32.4%	93.6%	173	$P_0 \geq 0.002, \alpha \leq 0.0731, \eta \geq 0.7753, V_0 \geq 10178.1503,$ $\delta \leq 0.0519$
37	31.2%	94.5%	165	$P_0 \geq 0.002, \alpha \leq 0.0731, \eta \geq 0.8017, V_0 \geq 10178.1503,$ $\delta \leq 0.0519$
38	30.4%	96.8%	157	$P_0 \geq 0.002, \alpha \leq 0.0731, \eta \geq 0.8217, V_0 \geq 10178.1503,$ $\delta \leq 0.0519$
39	29.2%	97.3%	150	$P_0 \geq 0.002, \alpha \leq 0.0731, \eta \geq 0.8385, V_0 \geq 10178.1503,$ $\delta \leq 0.0519$
40	28.0%	97.9%	143	$P_0 \geq 0.002, \alpha \leq 0.0731, \eta \geq 0.8532, V_0 \geq 10178.1503,$ $\delta \leq 0.0519$
41	26.8%	98.5%	136	$P_0 \geq 0.002, \alpha \leq 0.0731, \eta \geq 0.8679, V_0 \geq 10178.1503,$ $\delta \leq 0.0519$

```

let
    coverages = [b.coverage for b in cost_trajectory]
    densities = [b.density for b in cost_trajectory]

    fig = Figure(; size=(600, 400))
    ax = Axis(
        fig[1, 1];
        xlabel="Density",
        ylabel="Coverage",
        title="PRIM Peeling Trajectory (Cost-Based Failure)",
    )
    scatterlines!(ax, densities, coverages; markersize=8, linewidth=2, color=:orange)
    text!(ax, densities[1], coverages[1]; text="Start", align=:right, :bottom),
    fontsize=12)
    text!(
        ax,
        densities[end],
        coverages[end];
        text="End",
        align=:left, :top),
        fontsize=12,
    )

```

```
fig
end
```

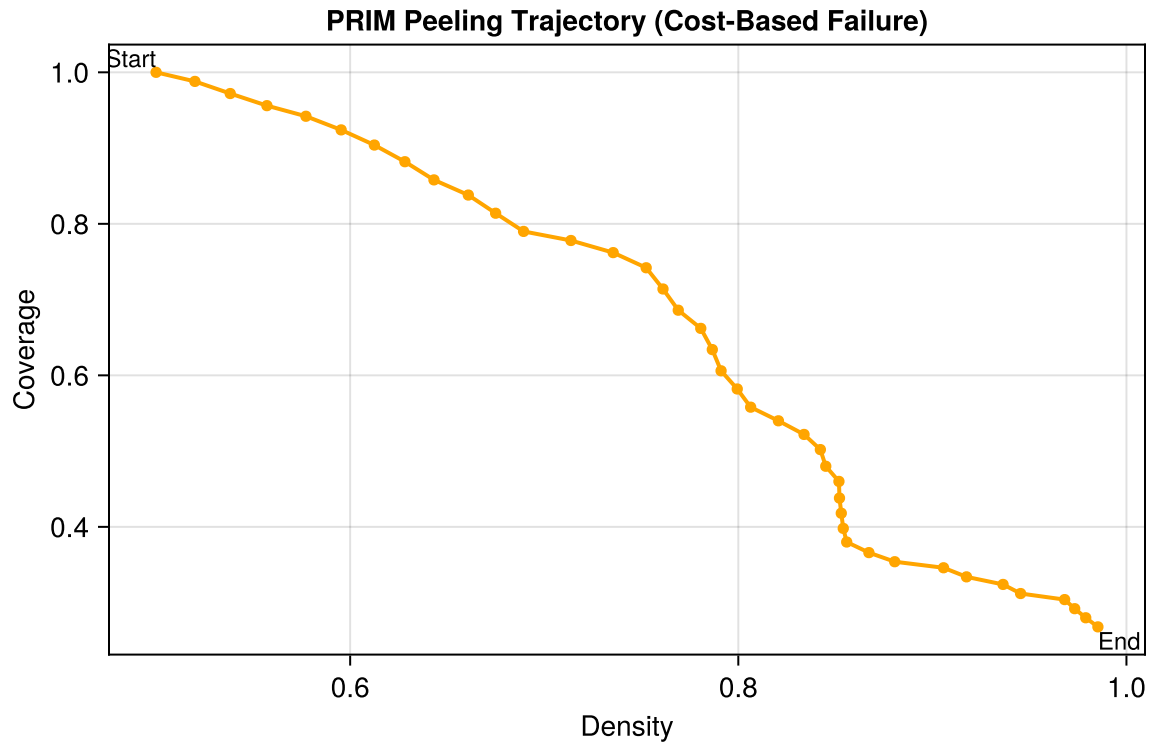


Figure 5: PRIM peeling trajectory for cost-based failure. Compare this to Figure 3 — the restricted dimensions may be different.

```
good_idx_cost = findfirst(b -> b.density > 0.7, cost_trajectory)
if good_idx_cost === nothing
    good_idx_cost = length(cost_trajectory)
end
box_cost = cost_trajectory[good_idx_cost]
println("Selected cost-based box $(good_idx_cost):")
println(" Coverage: $(round(100 * box_cost.coverage; digits=1))%")
println(" Density:  $(round(100 * box_cost.density; digits=1))%")
println(" Support:  $(box_cost.support) scenarios")
println()
initial_cost = cost_trajectory[1]
for (j, name) in enumerate(param_names)
    lo_changed = box_cost.lower[j] > initial_cost.lower[j]
    hi_changed = box_cost.upper[j] < initial_cost.upper[j]
    if lo_changed || hi_changed
        println("      $name:  [$(round(box_cost.lower[j]; sigdigits=3)),
$(round(box_cost.upper[j]; sigdigits=3))]")
    end
end
```

```
end
end
```

Selected cost-based box 13:

Coverage: 77.8%
Density: 71.4%
Support: 545 scenarios

P_0 : [0.00086, 0.004]
 α : [0.0251, 0.0731]
 η : [0.588, 1.2]
 V_0 : [6070.0, 25000.0]
 δ : [0.02, 0.055]

Your Response — Question 3

Compare the PRIM results for probability-based failure (Figure 3) and cost-based failure (Figure 5).

1. *Which parameters drive failure under each definition?*
2. *Are there parameters that matter for one definition but not the other? Why?*
3. *What does this tell you about the role of the analyst in scenario discovery?*

6 Wrap-Up

Key takeaways:

1. **Scenario discovery asks a different question:** given a fixed policy, under what conditions does it fail?
2. **PRIM finds interpretable boxes:** “failure occurs when $P_0 > X$ AND $\alpha > Y$ ” is a concrete, actionable statement.
3. **The failure definition matters:** probability-based and cost-based failures surface different uncertainties. The choice is a value judgment.
4. **Not all uncertainties matter equally:** PRIM reveals which inputs drive failure — and which are irrelevant. This focuses attention on the uncertainties worth reducing.

7 Submission

1. Write your answers in the response boxes above.
2. **Render to PDF:**
 - In VS Code: Open the command palette (Cmd+Shift+P / Ctrl+Shift+P) → “Quarto: Render Document” → select Typst PDF
 - Or from the terminal: `quarto render index.qmd --to typst`
3. **Submit the PDF** to the Lab 8 assignment on Canvas.
4. **Push your code** to GitHub (for backup):

```
git add -A && git commit -m "Lab 8 complete" && git push
```

Checklist

Before submitting:

- ☐ Packages load without error
- ☐ Ensemble runs successfully (1,000 scenarios)
- ☐ Default policy trajectories plotted (Figure 1)
- ☐ Question 1 answered (your policy choice)
- ☐ Your policy trajectories plotted (Figure 2)
- ☐ Probability-based PRIM run and plotted (Figure 3)
- ☐ Question 2 answered (interpret a PRIM box)
- ☐ Cost distribution plotted (Figure 4)
- ☐ Cost-based PRIM run and plotted (Figure 5)
- ☐ Question 3 answered (compare failure definitions)
- ☐ Notebook renders to PDF without errors

Bibliography

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- [2] J. H. Friedman and N. I. Fisher, “Bump Hunting in High-Dimensional Data,” *Statistics and Computing*, vol. 9, no. 2, pp. 123–143, 1999, doi: 10.1023/A:1008894516817.